

Introduction to Machine Learning

A short course used as a sample document for this RAG application

Chapter 1: What is Machine Learning

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn from data, without being explicitly programmed for every single task. Instead of writing fixed rules of the form "if X, then Y", we provide the algorithm with many examples, and it discovers the underlying patterns in the data on its own.

There are three main categories of machine learning: supervised learning, unsupervised learning, and reinforcement learning. Each category is suited to different types of problems.

1.1 Supervised Learning

In supervised learning, the algorithm receives labeled data — meaning that for every training example, the correct answer is already known. For example, a model that recognizes cats in pictures is trained with thousands of images, each labeled either "cat" or "not a cat". The goal of the model is to generalize, so that it can recognize cats in new images it has never seen before.

Two common tasks in supervised learning are classification (predicting a category, as in the cat example) and regression (predicting a numeric value, such as the price of a house based on its size and location).

1.2 Unsupervised Learning

Here the algorithm receives unlabeled data and must discover hidden structures or patterns on its own. A typical example is clustering: grouping a store's customers based on their purchasing behavior, without knowing in advance how many groups exist or what characteristics they share.

Chapter 2: Neural Networks

A neural network is a mathematical model loosely inspired by how biological neurons work. It is made up of layers of "nodes" (called artificial neurons), connected to each other through numerical weights.

Data enters through the input layer, passes through one or more hidden layers, where it is repeatedly transformed through multiplications and activation functions, and exits through the output layer as a final prediction.

2.1 Training a Network

Training means gradually adjusting the network's weights so that its predictions get closer and closer to the correct answers. This process uses two steps repeated many times: the forward pass, where data flows through the network and produces a prediction, and backpropagation, where the error between the prediction and the correct answer is used to correct the weights.

A popular optimizer for this process is Adam, which automatically adjusts the learning rate for each weight individually, usually leading to faster and more stable training compared to classic gradient descent.

Chapter 3: Embeddings and Semantic Search

An embedding is a numeric representation of a piece of text (a word, a sentence, or a paragraph) in the form of a vector — a sequence of numbers. Texts with similar meaning have vectors that are close to each other in this numeric space, even if they use different words.

This property enables semantic search: instead of looking for an exact match of certain words, we search for texts whose meaning is close to the question being asked, by comparing their vectors. The distance between two vectors is usually measured using cosine similarity, which indicates how close the directions of the two vectors are in space.

3.1 RAG (Retrieval-Augmented Generation)

RAG is an architecture in which a language model (LLM) does not answer purely from what it memorized during training, but also receives relevant context, retrieved from an external database, right before generating its answer. The typical steps are: splitting documents into small pieces (chunking), turning each piece into an embedding, storing them in a vector database, retrieving the most relevant pieces for a given question, and sending them to the LLM together with the question.

The main advantage of RAG is that the model can answer correctly about information it has never seen during training, such as the content of a document uploaded by the user themselves.